In [1]:
```
t1 = ()
```

In [2]:
```
t2 =(10,20,30)
```

In [3]:
```
t3 = (10.77,30.88,50.99)
```

In [4]:
```
t4 = ('one','two','three')
```

In [5]:
```
t5 = ('Aira',[10,20,30],25,(25,35),(12,26))
```

In [6]:
```
t6 = (100,'Aira','15.678')
```

In [7]:
```
t7 = ('aira',30,[50,100],[150,90],{'john','David'},(99,200))
```

Indexing

In [9]:
```
t2[0]
```

Out[9]:
```
10
```

In [10]:
```
t5[-1]
```

Out[10]:
```
(12, 26)
```

Slicing

In [11]:
```
t = (0,1,2,3,4,5,6,7,8,9)
```

In [12]:
```
t[:3]# returns first 3 vlues
```

Out[12]:
```
(0, 1, 2)
```

In [13]:
```
t[0:8] # returns 0 till 7 th index
```

Out[13]:
```
(0, 1, 2, 3, 4, 5, 6, 7)
```

In [14]:
```
t[:]
```

Out[14]:
```
(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
```

In [15]:
```
t[::]
```

Out[15]:
```
(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
```

In [16]:
```
t[-1]
```

Out[16]:
```
9
```

In [17]:
```
t[-10:-3]
```

Out[17]:
```
(0, 1, 2, 3, 4, 5, 6)
```

In [18]:
```
t[-3:-7]
```

Out[18]:
```
()
```

In [19]:
```
t[9:1]
```

Out[19]:
```
()
```

In [20]:
```
del t[0] #deleting one element is not possible in tuple
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[20], line 1
----> 1 del t[0]

TypeError: 'tuple' object doesn't support item deletion
```

In [21]:
```
t[0]= 100 #replacing the one element is not possible
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[21], line 1
----> 1 t[0]= 100

TypeError: 'tuple' object does not support item assignment
```

In [22]:
```
del t # deleting entire tuple is possible
```

Loop through a tuple

In [23]:
```
t = ('one','two','three','four','five','six','seven','eight','nine','ten')
```

In [24]:
```
for i in t:
    print(i)
```

```
one
two
three
four
five
six
seven
eight
nine
ten
```

In [26]:
```
for i in enumerate(t):
    print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
(8, 'nine')
(9, 'ten')
```

In [27]:
```
t
```

Out[27]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'te
n')
```

Membership

In [28]:
```
'one' in t
```

Out[28]:
```
True
```

In [29]:
```
'Eleven' in t
```

Out[29]:
```
False
```

In [30]:
```
if 'four' in t:
    print('Four in tuple')
else:
    print('Four not in tuple')
```

```
Four in tuple
```

In [31]:
```
if 'Eleven' in t:
    print('Eleven in tuple')
else:
    print('Eleven not in tuple')
```

```
Eleven not in tuple
```

Index Position

In [32]:
```
t.index('one')
```

Out[32]:
```
0
```

In [33]:
```
t.index('five')
```

Out[33]:
```
4
```

In [34]:
```
t.index('ten')
```

Out[34]:
```
9
```

In [35]:
```
t10 = (22,88,44,9,100,6,1,33,33)
```

In [36]:
```
t10.sort()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[36], line 1
----> 1 t10.sort()

AttributeError: 'tuple' object has no attribute 'sort'
```

In [38]:
```
sorted(t10)
```

Out[38]:
```
[1, 6, 9, 22, 33, 33, 44, 88, 100]
```

In [39]:
```
sorted(t10,reverse= True)
```

Out[39]:
```
[100, 88, 44, 33, 33, 22, 9, 6, 1]
```

Set

In [40]:
```
s = set()
```

In [41]:
```
type(s)
```

Out[41]:
```
set
```

In [43]:
```
s1= {1,2,3}
```

In [44]:
```
s1
```

Out[44]:
```
{1, 2, 3}
```

In [45]:
```
type(s1)
```

Out[45]:
```
set
```

In [46]:
```
s2={100,3,25,76,45,20,50}
s2
```

Out[46]:
```
{3, 20, 25, 45, 50, 76, 100}
```

In [47]:
```
s3 ={10+15j,9.89,14}
s3
```

Out[47]:
```
{(10+15j), 14, 9.89}
```

In [48]:
```
print(s)
print(s1)
print(s2)
```

```
set()
{1, 2, 3}
{50, 3, 100, 20, 25, 76, 45}
```

In [49]:
```
s2.add(35)
```

In [50]:
```
s2
```

Out[50]:
```
{3, 20, 25, 35, 45, 50, 76, 100}
```

In [51]:
```
s3 = s2.copy()
```

In [52]:
```
s3
```

Out[52]:
```
{3, 20, 25, 35, 45, 50, 76, 100}
```

In [53]:
```
print(s2)
print(s3)
```

```
{50, 3, 100, 20, 35, 25, 76, 45}
{3, 100, 35, 76, 45, 50, 20, 25}
```

In [54]:
```
s2[2]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[54], line 1
----> 1 s2[2]

TypeError: 'set' object is not subscriptable
```

In [55]:
```
s2.pop()
```

Out[55]:
```
50
```

In [56]:
```
s2
```

Out[56]:
```
{3, 20, 25, 35, 45, 76, 100}
```

In [57]:
```
s2.pop()
```

Out[57]:
```
3
```

In [58]:
```
s2
```

Out[58]:
```
{20, 25, 35, 45, 76, 100}
```

In [59]:
```
s2.remove()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[59], line 1
----> 1 s2.remove()

TypeError: set.remove() takes exactly one argument (0 given)
```

In [60]:
```
s2.remove(100)
```

In [61]:
```
s2.discard(100)
```

In [62]:
```
s2
```

Out[62]:
```
{20, 25, 35, 45, 76}
```

In [63]:
```
s2.remove(45)
```

In [64]:
```
s2
```

Out[64]:
```
{20, 25, 35, 76}
```

In [66]:
```
s2.discard(45) #doesn't throughs an error even if it is not in the set
```

In [67]:
```
s2.remove(45) #throughs an error if the element is not in the set
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[67], line 1
----> 1 s2.remove(45)

KeyError: 45
```

In [68]:
```
s2
```

Out[68]:
```
{20, 25, 35, 76}
```

In [69]:
```
s2[2]= 9
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[69], line 1
----> 1 s2[2]= 9

TypeError: 'set' object does not support item assignment
```

In [ ]: